4 a single lamp (F) = 4.8 A,

two lamps in parallel = 2.4 A (i.e. half through each),

three lamps in parallel = 1.6 A (i.e. one-third through each)

b potential difference = sum of series potential differences = (6.6 + 3.3 + 2.2) V = 12.1 V

5 Cells in series have a higher total potential difference than one cell, but will last for the same time as one cell. Cells in parallel will have the same potential difference as one cell but will last for a longer time than one cell.

6 a a = b = c = 0.1 A, d = 3 × a (or b or c) = 0.3 A, e = d + a (or b or c) = 0.4 A

b V across a = V across b = V across c = 0.6 V

c V across d = V across a + V across b + V across c = 1.8 V

V across E = 2.4 V

Battery potential difference = 1.8 V + 2.4 V = 4.2 V